

Habitat disturbance, not predation, is all that is required to influence habitat choice in juvenile snakes: A rejoinder to Lill

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Many animals respond to habitat disturbance by changing their behaviours in ways that minimize the risk of predation (Lima 1998; Gill *et al.* 2001). Although most research on this topic has focused on the effects of human disturbance on wildlife, native vertebrate species that disturb habitats can also influence habitat selection and behaviour of sympatric taxa (Lytle 2001; Frid & Dill 2002). Recently, we showed that animals disturbed 12.6% of 900 experimental rocks placed on nine transects across three plateaus in Morton National Park, New South Wales. During the 6-week study, animals disturbed rocks on soil substrates twice as frequently as rocks lying on rock substrates. As predicted by disturbance-risk theory (Frid & Dill 2002), juvenile snakes rarely used rocks on soil substrates as diurnal retreat sites, even though such rocks were readily available and conferred significant thermal benefits to snakes (Webb & Whiting 2006).

We concluded that superb lyrebirds were ‘the major agents of rock disturbance’ because 64 of 113 disturbed rocks (56.7%) showed the following evidence of lyrebird disturbance: (i) the leaf litter and soil in the location where the rock once lay (prior to being displaced) had been extensively raked in a manner similar to lyrebird scratchings; (ii) the snake models had been attacked, and contained beak imprints which were similar in size and shape; and/or (iii) both of these events had occurred. We were confident that lyrebirds had disturbed rocks because they were abundant on our study sites (large numbers of display mounds were present), and one of us (J. Webb) has seen lyrebirds disturbing small stones. Moreover, previous studies have reported that lyrebirds displace large quantities of leaf litter and soil during their foraging bouts (Lill 1996; Ashton & Bassett 1997) and can displace rocks up to 2 kg in weight (Adamson *et al.* 1983). Finally, we found no hoof prints or scats from feral pigs, rabbits or goats on the study sites, and we know of no other birds that could rake rocks sideways while foraging. The

brush-turkey, the only other bird that forages by raking leaf litter, has not been recorded from our study sites (Göth *et al.* 2006) nor has J. Webb observed it during 14 years of fieldwork.

The fact that we did not observe lyrebirds moving stones or attacking live snakes has drawn criticism. Alan Lill’s (this issue) primary objection to our study is that we ‘claim that superb lyrebirds *Menura novae-hollandiae* preyed on (or at least attacked) plasticine models of juveniles of two snake species . . . placed beneath rocks in the Morton National Park, New South Wales and which the birds discovered while foraging’. Lill objected to the indirect evidence we provided for this claim, and we respond to these objections below. First, it is true that we did not observe lyrebirds disturbing rocks in this study (though one of us has previously), but as pointed out above, we had good evidence to support our claim, and even Lill admits that lyrebirds occasionally disturb rocks while foraging (Lill 2006). Second, Lill pointed out that we did not compare beak imprints on our models with those made by a taxidermic specimen or a living lyrebird. This is a fair criticism. To address this issue, we obtained plasticine beak imprints from two preserved adult lyrebirds from the collection of the Macleay Museum, and compared the size and shape of these imprints to indentations on a subset of nine models that we retained from the field for identification purposes. The imprints on these models (shown in figures 4a,b, Webb & Whiting 2006) were similar in size and shape to the imprints we obtained from the preserved adult lyrebirds, which suggests that lyrebirds had pecked these models. However, because we could not accurately identify the avian species responsible for attacks on all of the plasticine models in the field (some models were bitten in half, or lacked complete beak imprints), we used multiple lines of evidence (above) to infer that lyrebirds had disturbed the rocks.

Finally, Lill says that even if lyrebirds attack juvenile snakes, this may be because they resemble earthworms and that it is not clear that lyrebirds attack juvenile snakes ‘frequently’. Lill ends by stating that it would be

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Accepted for publication July 2006.

a pity if our 'inadequately proven claim' that lyrebirds have played a role in the evolution of snake habitat choice becomes 'uncritically enshrined in the lyrebird literature'. We reiterate that the primary focus of our study was whether biotic disturbance (not predation) might influence habitat selection by small snakes. Second, we did not state that lyrebirds 'frequently' attack snakes. In our introduction we pointed out that lyrebirds eat soil invertebrates, and later, after discovering lyrebird disturbance to small rocks and attacks on snake models, we suggested that lyrebirds might eat small ectothermic vertebrates that live under stones. Lill suggests that small snakes may resemble large earthworms and this may invite mistaken attack, and we agree that this is certainly a possibility.

Irrespective of whether lyrebirds actually eat snakes, in the event that a lyrebird shifts a rock and unexpectedly exposes a small snake, there are three possible outcomes: (i) the bird ignores the snake (or does not see it); (ii) the bird attacks, but does not eat the snake and the snake survives; and (iii) the bird kills the snake. Irrespective of which outcome occurs, there is a significant cost to the snake because it loses the protection of its shelter site and becomes exposed to other avian predators (kookaburras, eagles, kites, etc.) that attack snakes (Webb & Whiting 2005). Thus, even if lyrebirds (and other mammals that disturb rocks) do not attack snakes, biotic disturbance to snake retreat-sites should eventually result in snakes shifting their habitat-use towards less disturbed habitats (e.g. Losos *et al.* 2004). Therefore, our general conclusions are not dependent on lyrebirds eating or attacking snakes, but simply on biotic disturbance to snake habitats. Although we used indirect evidence to implicate lyrebirds as one of the agents of disturbance, we suggest that discovering extensive scratch marks around a displaced rock, and lyrebird beak marks on the snake model, is akin to finding an injured pedestrian lying on a road beside tyre marks. In both cases, the inferential evidence strongly suggests the 'predator' responsible for the disturbance, but it cannot identify the individual responsible, nor can it determine how prevalent such individuals are in the population. These latter questions could be addressed by detailed studies of lyrebird behaviour, but they were beyond the scope of our study. We hope that this debate, and the results of our previous studies (Webb & Whiting 2005, 2006), will stimulate further research on the foraging ecology and diets of

lyrebirds, and their potential impacts on the biodiversity of forest leaf litter faunas.

ACKNOWLEDGEMENTS

We thank Allan Lill for his constructive criticism on our earlier paper, and for suggesting that we compare the beak imprints on our snake models with a taxidermic specimen of a lyrebird. We thank Margaret Humphrey for kindly allowing us to obtain plasticine beak imprints from the preserved lyrebirds under her care in the Macleay Museum.

REFERENCES

- Adamson D., Selkirk P. M. & Mitchell P. (1983) The role of fire and lyrebirds in the Sydney Basin. In: *Aspects of Australian Sandstone Landscapes* (eds R. W. Young & G. G. Nanson) pp. 81–93. Australian and New Zealand Geomorphology Group Special Publication 1, Wollongong.
- Ashton D. H. & Bassett O. D. (1997) The effects of foraging by the superb lyrebird (*Menura novaehollandiae*) in *Eucalyptus regnans* forests at Beenak, Victoria. *Aust. J. Ecol.* **22**, 383–94.
- Frid A. & Dill L. (2002) Human-caused disturbance stimuli as a form of predation risk. *Conserv. Ecol.* **6**, 11.
- Gill J. A., Norris K. & Sutherland W. J. (2001) Why behavioural responses may not reflect the population consequences of human disturbance. *Biol. Conserv.* **97**, 265–8.
- Göth A., Nicol K. P., Ross G. & Shields J. (2006) Present and past distribution of Australian brush-turkeys *Alectura lathami* in New South Wales – implications for management. *Pac. Conserv. Biol.* **12**, 22–30.
- Lill A. (1996) Foraging behavioural ecology of the Superb Lyrebird. *Corella* **20**, 77–87.
- Lill A. (2006) Lyrebirds and juvenile snakes, a comment on Webb and Whiting. *Austral Ecol.* **31**, 904.
- Lima S. L. (1998) Stress and decision making under the risk of predation: recent developments from behavioural, reproductive, and ecological perspectives. *Adv. Study Behav.* **27**, 215–90.
- Losos J. B., Schoener T. W. & Spiller D. A. (2004) Predator-induced behaviour shifts and natural selection in field experimental lizard populations. *Nature* **432**, 505–8.
- Lytle D. A. (2001) Disturbance regimes and life-history evolution. *Am. Nat.* **157**, 525–36.
- Webb J. K. & Whiting M. J. (2005) Why don't small snakes bask? Juvenile broad-headed snakes trade thermal benefits for safety. *Oikos* **110**, 515–22.
- Webb J. K. & Whiting M. J. (2006) Does rock disturbance by superb lyrebirds (*Menura novaehollandiae*) influence habitat selection by juvenile snakes? *Austral Ecol.* **31**, 58–67.